CCTV PLACEMENT ASSESSMENT

Residential Property -- Demo Scan (4_25_2026)

Analysed by HawkEye Sentinel AI | Groq Llama 3 | Importance-Weighted Optimizer

Report Date: May 07, 2026 | Classification: CONFIDENTIAL

1. EXECUTIVE SUMMARY

HawkEye Sentinel performed an AI-driven security assessment of the scanned residential property. Using spatial reasoning (Groq Llama-3.3-70b), importance scoring, and an importance-weighted greedy placement optimizer, the system identified optimal camera positions that maximize coverage of high-priority zones (corridors, entry points) while staying within the defined budget.

The assessment identified 4 rooms, 8 entry points (doors/windows), and generated adversarial threat paths for two attacker profiles (Burglar and Pro). The recommended 7-camera layout achieves 100% coverage of all entry points and 92.4% floor coverage at a total cost of Rs. 1,35,500.

2. ASSESSMENT METRICS

Property	**Residential -- Demo Scan (Polycam 4_25_2026)**
Floor Area	**84.22 sq.m.**
Rooms Analysed	**4 (Entry Room, Corridor x2, Office/Living)**
Entry Points	**8 (Doors x6 + Windows x2)**
Main Entrance	**Door_5 (width 1.8m, highest priority)**
Recommended Cameras	**7**
Camera Coverage	**92.4% of floor area**
Entry Points Covered	**8 / 8 (100%)**
Blind Spots	**2 (mitigable within budget)**
Total Cost	**Rs. 1,35,500 (Rs. 74,500 under budget)**
Budget	**Rs. 2,10,000**
Analysis Source	**Groq Llama-3.3-70b + Geometry Prior**

3. RECOMMENDED CAMERA PLACEMENTS

Camera	Type	Zone	Cost	Rationale	Priority
CAM-01	Bullet 2K	Entry / Front Door Zone	Rs.17,000	Entry priority -- delta coverage +28.4%	**HIGH**
CAM-02	Dome WDR	Main Corridor (West)	Rs.23,000	Corridor pivot -- delta coverage +21.7%, +1 entry	**HIGH**
CAM-03	Dome IR	NW Staircase / Corner	Rs.19,000	Insider-threat bias -- NW blind spot	**MEDIUM**
CAM-04	Dome 4K	Living / Office Room	Rs.29,000	Wide-angle coverage of main living space	**MEDIUM**
CAM-05	Dome WDR	East Corridor	Rs.23,000	Secondary corridor coverage	**MEDIUM**

CAM-06	Bullet 2K	Back Door / Window Zone	Rs.17,000	Rear entry point coverage	**MEDIUM**
CAM-07	Dome IR	Hallway Junction	Rs.19,000	IR coverage for low-light corridor	**LOW**

4. AI IMPORTANCE SCORING (GROQ LLAMA-3.3-70B)

Room ID	Inferred Type	Score	Reason
Room_0	Entry Room / General Living	**0.50**	Wide rectangle with single door, connected to corridor.
Room_1	Corridor / Hallway	**0.80**	Long narrow shape (7.1:1 aspect), multiple doors.
Room_2	Corridor / Hallway	**0.85**	Long narrow, 3 doors, some furniture -- key choke point.
Room_3	Living Room / Office	**0.55**	Wide rectangle with chairs, general use space.

5. ADVERSARIAL THREAT PATH ANALYSIS

HawkEye Sentinel ran A* adversarial pathfinding for two threat profiles from all 8 entry points to the highest-priority target room. The algorithm routes attackers through blind spots, penalising camera-visible zones.

BURGLAR	Avoids cameras moderately. Fastest exit route. 3 breach cameras on primary path.
PRO	Maximum stealth. Slowest, most dangerous. Routes through blind spots. 1 breach camera.

6. RECOMMENDATIONS

1. Install CAM-01 at the main entrance (Door_5) immediately -- highest importance score (0.98).
2. Prioritise corridor coverage: Room_1 and Room_2 are key choke points at 0.80 and 0.85 scores.
3. Address 2 identified blind spots in the NW corner using CAM-03 (Dome IR) for night visibility.
4. Consider upgrading CAM-04 to a PTZ unit (Rs. 50,000) for pan coverage of the living/office area.
5. Schedule periodic re-scans when furniture layout changes -- scores are geometry-dependent.
6. Enable lighting simulation to verify HDR/WDR selection for the SW corridor glare window (16:00-18:00).